Readmission Prevention Worklist — Recommended Operating Pattern

Draft v0.4 • November 2, 2026

Eight operational steps from the nightly EHR extract to care manager outreach and measured 30-day outcomes, with the tables and assets touched in each step.

Deployment: workspace <env>_clinical_analytics • catalog <env>_care_management (dev: rtr_demo_catalog) • Lakebase project readmission-worklist, database readmission_worklist

1 CONFIGURE PROGRAM

Care Management Leadership

Adjust risk-tier cut-offs, outreach SLAs, and care manager capacity in the app's Program Settings tab; seed files provide the initial load.

- **Program Settings tab** (SIGN IN)
- **Lakebase readmission_worklist_app.program_setting_change** (APPEND)
- **Seed: facility, risk_tier_policy, care_team_assignment** (SEED)

2 INGEST + CURATE

`<env>_readmit_daily_cycle › ingest_curate`

Auto Loader picks up new EHR extracts, expectations quarantine bad rows, app changes are applied, and features plus 30-day labels refresh.

- **ehr_raw.ehr_landing** (READ)
- **ehr_bronze.*_raw** (APPEND)
- **clinical_silver.patient, encounter, diagnosis, follow_up_appointment, facility** (UPSERT)
- **readmission_worklist_app_bronze.lb_*_history** (READ)
- **care_management_silver.outreach_activity, risk_tier_policy, care_team_assignment, app_audit_event** (UPSERT)
- **readmission_worklist_gold.discharge_features, readmission_outcome** (REFRESH)

3 SCORE + PRIORITIZE

`<env>_readmit_daily_cycle › score_discharges`

Score yesterday's discharges with the @champion model, explain the top three risk drivers, and assign a risk tier.

- **readmission_worklist_gold.readmission_risk@champion** (LOAD)
- **readmission_worklist_gold.discharge_features** (READ)
- **care_management_silver.risk_tier_policy** (READ)
- **readmission_worklist_gold.discharge_risk_score** (APPEND)

4 BUILD + PUBLISH WORKLIST

`<env>_readmit_daily_cycle › build_worklist`

Route high- and medium-risk discharges to the assigned care manager, draft an outreach brief with ai_query, and refresh Lakebase.

- **readmission_worklist_gold.discharge_risk_score** (READ)
- **care_management_silver.care_team_assignment** (READ)
- **Foundation Model endpoint** (GENERATE)
- **readmission_worklist_gold.care_worklist, care_team_roster** (MERGE)
- **Lakebase readmission_worklist_gold.*_synced** (SYNC)

5 WORK THE WORKLIST

Care Manager

Open today's prioritized list, review the drivers and brief, call the patient, and log the outcome and next step.

- **Worklist tab** (SIGN IN)
- **Lakebase readmission_worklist_gold.care_worklist_synced** (READ)
- **Lakebase readmission_worklist_app.outreach_activity** (INSERT)

6 SYNC CHANGES BACK

Lakebase CDF + Databricks Apps telemetry

Outreach and settings changes land as an append-only change log within about 15 seconds; audit events and request traces land through OpenTelemetry.

- **Lakebase readmission_worklist_app.*** (CDC)
- **readmission_worklist_app_bronze.lb_*_history** (APPEND)
- **App telemetry (OpenTelemetry)** (EMIT)
- **readmission_worklist_app_bronze.app_otel_logs, traces, metrics** (APPEND)

7 MEASURE + ASK GENIE

Care Management Leadership / Quality

Track reach rate, 7-day follow-up, and readmissions by tier and facility; ask follow-up questions in plain English.

- **Readmission Program Genie Agent** (ASK)
- **readmission_worklist_gold.readmission_program_metrics** (READ)
- **readmission_worklist_gold.readmission_outcome** (READ)
- **readmission_worklist_gold.care_worklist** (READ)

8 RETRAIN + REPEAT

`<env>_readmit_model_training`

Monthly: retrain on matured labels and promote the @challenger only if it beats @champion on AUC and calibration gates.

- **readmission_worklist_gold.discharge_features** (READ)
- **MLflow experiment** (LOG)
- **readmission_worklist_gold.readmission_risk@challenger** (REGISTER)
- **readmission_worklist_gold.readmission_risk@champion** (PROMOTE)

next daily cycle (retrain monthly)

READMISSION WORKLIST APP

AppKit app. Care managers work the Worklist tab; leadership uses the Program tab (metrics + embedded Genie Agent) and Program Settings. It reads synced gold tables, writes only to its Lakebase schema readmission_worklist_app, and emits audit events through app telemetry.

GOVERNANCE (EVERY STEP)

PHI columns carry governed tags. Tag-driven ABAC policies on silver and gold schemas mask direct identifiers and filter rows by facility. The CDF landing schema readmission_worklist_app_bronze stays unmasked (a CDF requirement), so access is limited to the pipeline identity. Telemetry carries IDs only, never PHI.

Ownership: red = Care Management Leadership • green = Care Manager • blue = automated Data Platform job or managed sync

Actions: READ = read only • INSERT = add rows • APPEND = append-only insert • SEED = initial load from version-controlled files • UPSERT = AUTO CDC apply • MERGE = keyed upsert • REFRESH = materialized view recompute • LOAD / REGISTER / PROMOTE = UC model alias ops

LOG = MLflow run • GENERATE = ai_query call • SYNC = Delta to Lakebase synced table • CDC = Lakebase CDF to Unity Catalog • EMIT = OpenTelemetry export • ASK = Genie question • SIGN IN = authenticated app session

Readmission Prevention Worklist — Environment-Local Deployment Pattern

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One Declarative Automation Bundle deploys every asset per environment; each workspace reads and writes only its own <env> catalog and Lakebase project.

Catalog Design Pattern

UNITY CATALOG CATALOG
`<env>_care_management` (DEV: RTR_DEMO_CATALOG)

READ/WRITE binding

- SCHEMA · SOURCE `ehr_raw`
  - `Volume: ehr_landing`
- SCHEMA · SOURCE `ehr_bronze`
  - `patient_raw` `encounter_raw` `diagnosis_raw` `appointment_raw` `quarantine_raw`
- SCHEMA · SOURCE (APP) `readmission_worklist_app_bronze`
  - `lb_outreach_activity_history` `lb_program_setting_change_history` `app_otel_logs` `app_otel_traces` `app_otel_metrics`
  - `CDF + app telemetry; restricted, no masks`
- SCHEMA · DOMAIN `clinical_silver`
  - `patient` `encounter` `diagnosis` `follow_up_appointment` `facility`
- SCHEMA · DOMAIN `care_management_silver`
  - `outreach_activity` `risk_tier_policy` `care_team_assignment` `program_setting` `app_audit_event`
- SCHEMA · DATA PRODUCT `readmission_worklist_gold`
  - `discharge_features` `discharge_risk_score` `care_worklist` `care_team_roster` `readmission_outcome` `readmission_program_metrics (metric view)` `Model: readmission_risk` `care_worklist_synced` `care_team_roster_synced`
- SCHEMA · SHARED `phi_governance`
  - `fn mask_direct_identifier` `fn mask_birth_date` `fn filter_by_facility` `ABAC policies on tag phi`

Naming convention

<source_system>_raw / _bronze (the app is a source system) • **<data_domain>_silver** • **<data_product>_gold**. Lakebase mirrors UC at the schema level: a synced gold schema keeps its UC name in Postgres, and the app's Postgres schema <app> and its telemetry land in UC as <app>_bronze. Synced tables take a _synced suffix.

PHI tagging rule

Every PHI column carries governed tag **phi = direct_identifier | quasi_identifier**; ABAC policies on silver and gold schemas apply masks and the facility row filter to new tables automatically. CDF landing schemas are excluded because CDF destinations must not carry masks or filters.

Minimum-necessary rule

Only worklist columns a care manager needs to make the call leave the lakehouse; Lakebase never holds diagnosis history or the full chart.

Workspace Design Pattern

DATABRICKS WORKSPACE
`<env>_clinical_analytics` (DEV: FEVM-CLASSIC-STABLE)

- LAKEFLOW Pipeline
  - `<env>_readmit_pipeline (serverless SDP)`
- LAKEFLOW Jobs
  - `<env>_readmit_daily_cycle: ingest_curate → score_discharges → build_worklist`
  - `<env>_readmit_model_training (monthly)`
  - `<env>_readmit_setup (synced tables + CDF + Genie, idempotent)`
  - `<env>_readmit_source_simulator (demo only)`
- LAKEBASE PROJECT `readmission-worklist`
  - `database readmission_worklist`
  - `schema readmission_worklist_gold (synced, read-only): care_worklist_synced, care_team_roster_synced`
  - `schema readmission_worklist_app (app-owned, REPLICA IDENTITY FULL): outreach_activity, program_setting_change`
- DATABRICKS APP `<env>-readmission-worklist`
  - `AppKit (Node/TypeScript/React)`
  - `resources: postgres, sql_warehouse, genie_space`
  - `telemetry_export_destinations: app_otel_*`
  - `Worklist tab` `Program tab` `Program Settings tab`
- GENIE AGENT Readmission Program
  - `readmission_worklist_gold.readmission_program_metrics`
  - `readmission_worklist_gold.readmission_outcome`
  - `readmission_worklist_gold.care_worklist`
- SERVING Foundation Model
  - `${var.llm_endpoint} (pay-per-token)`
  - `SQL warehouse <env>_readmit_wh`

Dev deviation

rtr_demo_catalog grants account users SELECT catalog-wide, which inherits into the unmasked CDF landing schema. Accepted for dev because all data is synthetic; prod uses a dedicated catalog with no account-wide grants.

App access rule

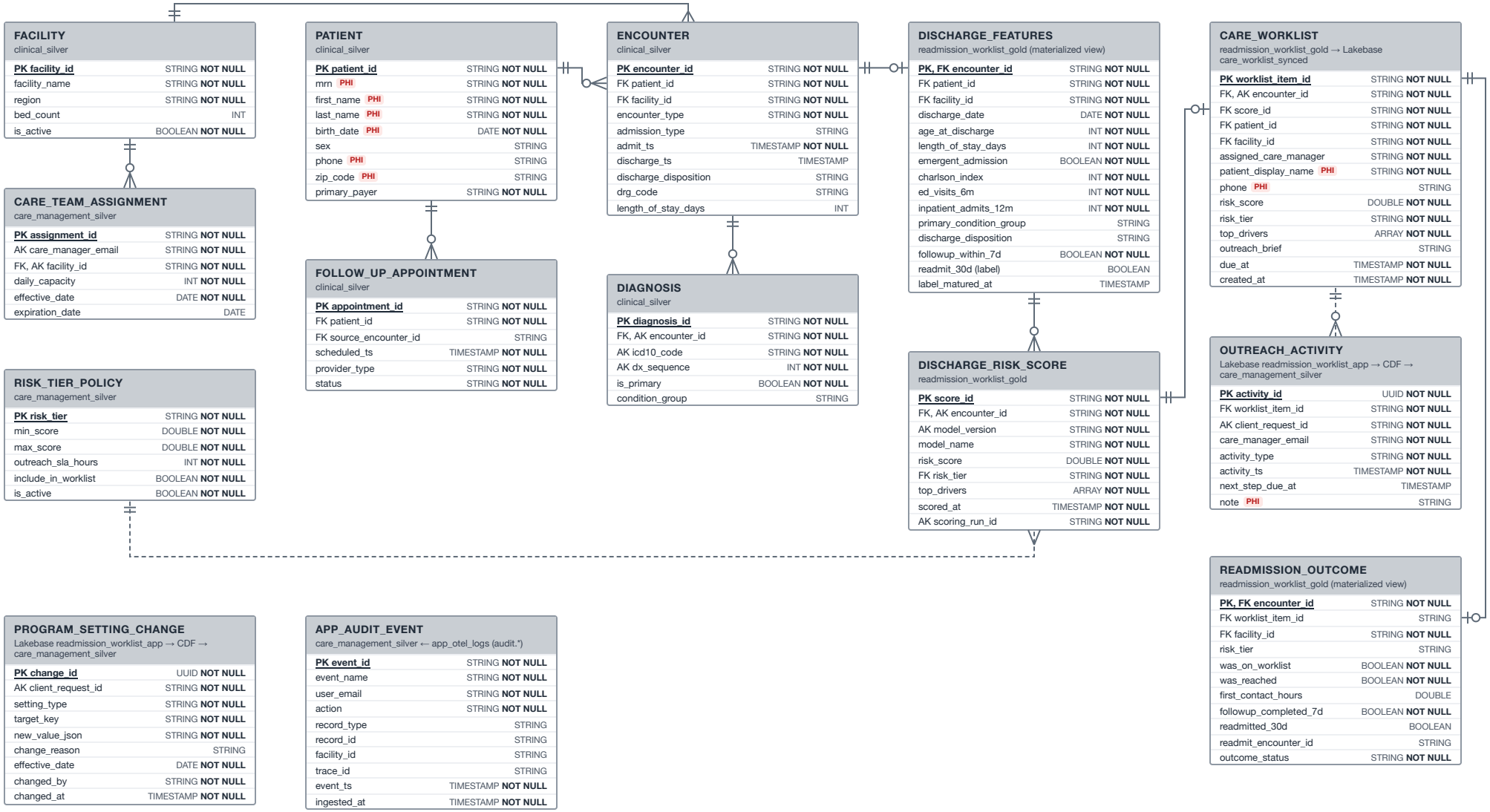
UC row filters do not reach Postgres, so every app query is scoped to the signed-in user's facility assignments; Program Settings requires the care_mgmt_leadership group; Genie runs on behalf of the user so UC policies apply; audit events carry IDs only, never PHI.

Environment guardrail

Bundle targets dev and prod set <env> and the catalog; jobs reject any catalog or Lakebase target that does not match the target's configured values.

Readmission Worklist Entity-Relationship Diagram

Tables deployed in <env>_care_management (dev: rtr_demo_catalog) and Lakebase database readmission_worklist; recommended logical keys and workflow relationships.



Status, tier, type, disposition, and payer columns are controlled STRINGs. PHI marks columns carrying governed tag phi; ABAC policies mask them for non-care-team readers.

PK/FK are Unity Catalog informational constraints (not enforced); required columns are NOT NULL; jobs enforce uniqueness, idempotency, and referential checks. AK columns are hashed to generate the PK.

Every silver and gold table also carries audit_created_at / _by / _process / _run_id and audit_updated_* columns (omitted above for readability). program_setting_change rows are validated by the pipeline and applied to risk_tier_policy or care_team_assignment by setting_type + target_key (logical, not drawn). app_audit_event is de-duplicated on event_id from app_otel_logs and carries IDs only.

readmission_program_metrics (metric view over readmission_outcome) defines readmission_rate, reach_rate, followup_7d_rate, and est_cost_avoided once, for Genie and the app.

Legend

- 1 to zero-or-many
- 1 to zero-or-one
- Logical reference (across Lakebase)

PK primary key • FK foreign key • AK alternate key